Levels of Measurement

Operationalizing and Categorizing: Variable Types

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Where We Left Off

Defining complex concepts:

- **Conception:** the mental image of the phenomenon.
- **Concepts:** social constructs made explicit and specific for the purpose of research
- **Conceptualization:** the process through which conceptions are made more precise
- Good concepts are observable and precise

Measuring complex concepts:

- **Operationalization:** the process of turning concepts into *operational* measures. The process of turning “squishy” concepts into ”firm,” consistent, comprehensible, measurable observations.

Good Concepts Are...

Observable: measurable with our ordinary senses.

- Directly observable: place of residence; weight; age.
- Indirectly observable: social status; regime type.
- Unobservable: human nature; divine intervention.

Precise: clear about the properties it summarizes; refers to only **one** set of properties of a given phenomenon.

- E.g.: "*level* of education" vs. "education."
- What are other dimensions of education?

The 3 Benchmarks of Measurement

1. **Precision**
   - The amount of information about the attributes or behavior measured contained in our measurement (*precision*).
2. **Reliability**
   - The extent to which a measurement procedure yields the same results on repeated trials (*consistency*).
3. **Validity**
   - The degree of correspondence between the measure and the concept it is thought to measure (*accuracy*).

Evaluating Validity

Face validity: does it measure the concept how we want to measure it?

- Example concept: worker morale.
- Measure 1: number of grievances worker files with the union.
- Measure 2: number of books worker checks out at the library.

Content validity: does it capture the full range of meanings of the concept it is supposed to measure?

- Example concept: math skills.
- Measure 1: test based on addition alone.
- Measure 2: test based on addition, multiplication, subtraction, division.

Levels of Measurement

Categorical Variables:

- Values correspond to different categories within the variable.
- The categories are exhaustive and mutually exclusive. (What does this mean?)

1. **Nominal Variables:**
   - **BUT**, the categories are not naturally ordered!
   - E.g.: Car make (Toyota, Ford, Honda, etc.); Colors (blue, green, red, etc.).
2. **Ordinal Variables:**
   - **BUT**, we know how to order the categories in some logical way!
   - E.g.: Ideology (Lib–Mod–Con); Height (short, average, tall, dos Santos).
3. **Dichotomous Variables:**
   - **BUT**, there are only two categories!
   - E.g.: On/Off; Present/Absent; Yes/No; Cool/Uncool.
   - Sometimes called *binary* variables or *dummy* variables, too.

Continuous Variables:

- Values correspond to different points within the variable's range.
- Values imply a certain order *and* distance among categories.
- Unlike ordinal variables, we know exactly how far apart observations are.
- E.g.: GDP growth (could be $-\infty$ to ∞); Temperature (could be $-\infty$ to ∞).

Interval/Ratio Variables:

- Same as continuous, but the values are bounded by nature.
- For our purposes, we will treat continuous and interval **as the same** level of measurement.
- E.g.: Age (You can't be -5 or 650 years old!); Height (You can't be negative inches tall!).

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology (Ordinal)
Political party

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology (Ordinal)
Political party (Nominal)
Support for a policy

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology (Ordinal)
Political party (Nominal)
Support for a policy (Dichotomous, Ordinal, Continuous/Interval)
Democracy

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology (Ordinal)
Political party (Nominal)
Support for a policy (Dichotomous, Ordinal, Continuous/Interval)
Democracy (Dichotomous, Ordinal, Continuous/Interval)
Year

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology (Ordinal)
Political party (Nominal)
Support for a policy (Dichotomous, Ordinal, Continuous/Interval)
Democracy (Dichotomous, Ordinal, Continuous/Interval)
Year (Continuous/Interval)
Probability of something

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology (Ordinal)
Political party (Nominal)
Support for a policy (Dichotomous, Ordinal, Continuous/Interval)
Democracy (Dichotomous, Ordinal, Continuous/Interval)
Year (Continuous/Interval)
Probability of something (Continuous/Interval)
Race

Practice

What levels of measurement are appropriate for these concepts?

(Consider how different arguments can be made depending on different units-of-analysis)

Country (Nominal)
Voter turnout (Continuous/Interval?)
Ideology (Ordinal)
Political party (Nominal)
Support for a policy (Dichotomous, Ordinal, Continuous/Interval)
Democracy (Dichotomous, Ordinal, Continuous/Interval)
Year (Continuous/Interval)
Probability of something (Continuous/Interval)
Race (Nominal)